

Stage 1 Report

1. Problem Statement

Football clubs invest significant resources recruiting talented players, but identifying high future potential is challenging since it depends on age, match performance, market value, transfer history, and international experience - and manual evaluation is time-consuming and subjective. This project develops an AI-Based Football Player Future Potential Rating Prediction system that predicts a player's Future Potential Rating (0-100) from historical statistics, assisting clubs, scouts, and agents in data-driven decisions.

2. Objectives

- Develop an AI model to predict a Football Player Future Potential Rating (0-100).
- Analyze historical football player statistics using machine learning.
- Identify important factors affecting future player potential.
- Assist clubs, scouts, and agents in evaluating players objectively.

3. Stakeholders and Intended Use Cases

- Football Clubs - estimate a player's future potential before making transfer decisions.
- Football Scouts - evaluate newly discovered players by entering performance statistics.
- Player Agents - use the predicted rating during transfer negotiations.

4. Why AI is Suitable

Predicting future performance requires analyzing many interacting factors simultaneously - performance, market value, career progression, and competition level - and these relationships do not follow a fixed rule; a player's stats can mean different things depending on age and league strength. Machine Learning can learn such complex, non-linear relationships from historical data, giving consistent and objective predictions that are difficult to obtain through manual analysis alone.

5. Dataset Identification

Dataset: Player Scores Dataset (Transfermarkt Football Data)

Source: Kaggle (<https://www.kaggle.com/datasets/davidcariboo/player-scores>)

Real-world football data from Transfermarkt is used. Multiple CSV files (Players, Appearances, Player Valuations, Transfers, Clubs, Competitions) are merged and preprocessed into a single player-level dataset for machine learning.

6. Input Columns

The model will use around 50 selected input features such as player profile, match performance, market value, transfer history, international experience, and club/competition details. The user will manually enter the required values.

7. Target Column

Target Variable: Player Future Potential Score (0–100), with the rating category derived as **Outperforming**, **Average**, or **Underperforming**.

8. How Will User Input Be Captured?

The end user (club, scout, or agent) enters player statistics through the application's input form. These values will be preprocessed and used by the model to predict the Future Potential Score.

Workflow:

User enters player's statistics -> Data Preprocessing -> PCA (if applicable) -> AI Model -> Future Potential Rating (0-100)

9. Data Source, Scope and Constraints

Source: Transfermarkt Player Scores dataset on Kaggle.

Scope: Predicting future potential of professional football players using historical statistics.

Constraints:

- Some player records may contain missing values.
- Player performance may be affected by injuries or external factors not in the dataset.
- Prediction quality depends on accuracy of user-entered statistics.
- The model supports, not replaces, professional scouting.

10. Conclusion

The proposed system will use real-world football statistics to estimate a player's future potential. By combining historical player data with machine learning techniques, the system aims to provide reliable and objective predictions that support football clubs, scouts, and player agents in making better recruitment and career-related decisions.